

What does it take to run a Linux distribution?

An overview of the infrastructure behind Arch Linux

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Meta

- **Description:** <https://cfp.tuebix.org/tuebix-2026/talk/YG8988/>
- **Slides:** <https://md.archlinux.org/p/2026-tuebix-presentation#/>
(interactive presentation)

What? — Scope

- Around 50 Servers and various services all around the 
 - Root servers / Cloud servers
 - Paid servers / sponsored servers
- Infrastructure as code repository 
gitlab.archlinux.org/archlinux/infrastructure
 - Terraform (2.6k LoC HCL)
(tf-stage1 / tf-stage2)
 - Ansible (15k LoC Jinja / 11k YAML)

Ansible

```
- name: Setup bugbuddy.archlinux.org
hosts: bugbuddy.archlinux.org
remote_user: root
roles:
  - { role: common }
  - { role: firewalld }
  - { role: wireguard }
  - { role: sshd }
  - { role: root_ssh }
  - { role: prometheus_exporters }
  - { role: promtail }
  - { role: fail2ban }
  - { role: bugbuddy }
```

```
- name: Install anubis bot policy dropins
copy: src={{ item }}.yaml dest=/etc/anubis/{{ item }}.yaml
notify: Restart anubis
when: anubis_extra_rules is defined
loop: "{{ anubis_extra_rules }}"
```

tf-stage1

```
machines = {  
    "accounts.archlinux.org" = {  
        server_type = "cx23"  
        domain      = "accounts"  
        location    = "hel1"  
    }  
    # ...  
}
```

```
archlinux_org_a_aaaa = {  
    "@" = {  
        ipv4_address = "209.126.35.79"  
        ipv6_address = "2604:cac0:a104:d::3"  
        ttl          = 60  
    }  
    archive = {  
        ipv4_address = "49.12.124.107"  
        ipv6_address = "2a01:4f8:242:5614::2"  
    }  
}
```

tf-stage2

```
resource "fastly_service_vcl" "fastly_mirror_pkgbuild_com" {  
  name = "Arch Linux Fastly Mirror"  
  
  domain {  
    name      = "fastly.mirror.pkgbuild.com"  
    comment   = "Arch Linux Fastly Mirror"  
  }  
  
  backend {  
    address      = "mirror.pkgbuild.com"  
    name         = "MIRRORPKGBUILD"  
    port         = 443  
    override_host = "mirror.pkgbuild.com"  
    use_ssl      = true  
  }  
}
```

Arch Linux Principles

1. **Simplicity**
2. **Modernity**
3. **Pragmatism**
4. **User centrality**
5. **Versatility**

→ *Keep It Simple Stupid (KISS)*

Additional Goals 🏆

- As much independence as possible 🗽
- Ensure easy Maintainability 🚧
- Maximize uptime 100
- Enable the other contributors in the project 🧠
- Contributing should be fun 😊
- Security & privacy-preserving measures 🔒

How? — Tech Stack

“Professional homelabbing”

- Servers with Arch Linux (btw!) 🤪
- Monitoring via Grafana / Prometheus / Loki / Tempo 🔍
- 1 Service → 1 VServer (mostly)
- NGinX as reverse proxy / TLS termination 🔄
- Borg for backup 💾
- BTRFS as filesystem 📁

Social challenges

- **Getting started:** New people and  & 
- **Culture:** money & companies 
- **Availability:** Timezones, no “shifts”, no fixed commitment 
- **Opinions:** Resistance to change & finding consensus 
- **Motivation:** Maintenance (“boring”)  vs. feature work 
- **Recognition:** infra admin is a thankless job 

Costs — What do we pay?

- **Cheap:** ~1k per Month for the overall infra (hosting at Hetzner) 
- ~~Cheap~~: Money is not the only resource that we have 
- **Free:** GeoMirrors, Fastly CDN, HAProxy ADN, Gitlab, Hetzner Discount, Statuspage 
- **Donations:** Although money is coming in, more  better 

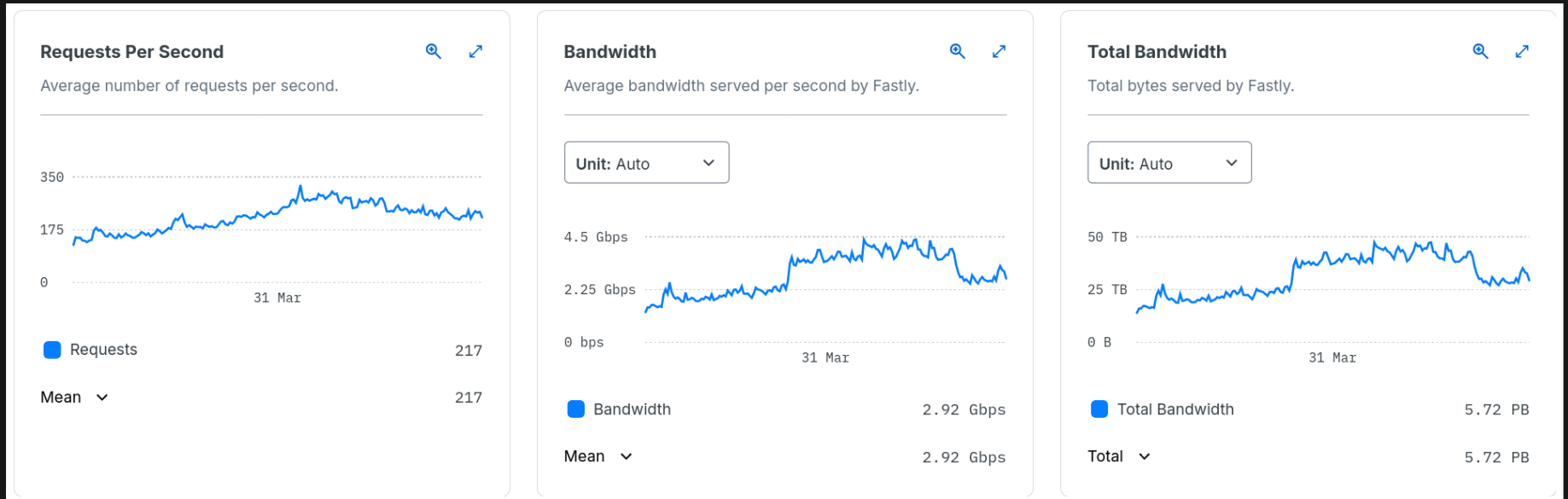
Service Catalog

- **Auth** : Keycloak for Single-Sign-On
- **Knowledge resources** : Wiki, manpages, static pages
- **building & running** : gitlab runners, repro, buildservices
- **Bots** : phrik, bugbuddy, gluebuddy, bumpbuddy

Service Catalog

- **Communication** : mailing lists, mailserver, matrix, mumble
- **Collaboration** : HedgeDoc, GitLab, security tracker, BBS, AUR
- **Package ecosystem** : repo server, mirror infra, archive, archweb

Fastly CDN



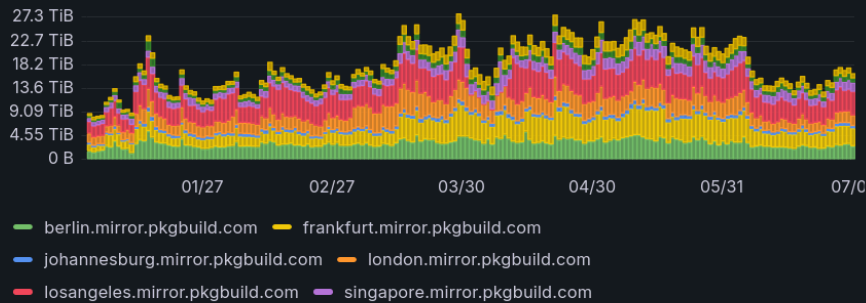
(2026-01 – 2026-07)



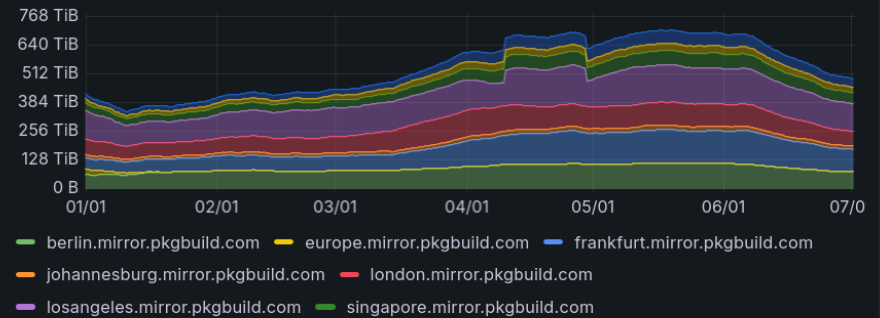
Cache consistency excursion?

GeoMirrors

Daily Outgoing Traffic (stacked)



Outgoing Traffic (stacked, 30 day rolling period)



(2026-01 – 2026-07)

Arch Linux User Repository (AUR) 🤠

- Scope: 108k Packages / 141k Users / 70 Admins
- **DISCLAIMER** (...)
- [AURweb](#)
- [goaurrpc](#)

debuginfod 🐛

- binary ➡ debug package ➡ debuginfod
- \$DEBUGINFOD_URLS
- valgrind, gdb, systing etc.

DRM Panic Screen



github.com/kdj0c/panic_report



Talk

Reproducible builds

<https://reproducible-builds.org/docs/definition/>

<https://reproducible.archlinux.org/>

to deal with —

- People scripting & Downstream distributions

- [ericc-ch/copilot-api#262](#) 
- Scraping   

to deal with —

- Moderating content (CSAM etc.)
- Cease-and-desist (IDE, sdks, emulators)
- Evaluating sponsorship contracts
- Age verification laws



to deal with —



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&



- Malware
- DDoS
- Credential leaks & hacked accounts
- Emergency patching & security embargos

Get involved

https://wiki.archlinux.org/title/Getting_involved

<https://whatcanidofor.archlinux.org/>

Q&A + Discussion

About me

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- Freelance MLOps / DevOps Consultant
- FOSS:
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 - Arch Linux
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